- $-A = (-1)A$
- $A + B = B + A$
- $A - B = A + (-1)B$
- $(A + B) + C = A + (B + C)$ where A, B and C have the same order.
- $A(BC) = (AB)C$ (ii) $A(B+C) = AB+AC$ (iii) $(A+B)C = AC+BC$
- The transpose of A, denoted by A^T is obtained by interchanging rows and columns of A.
  (i) $(A^T)^T = A$, (ii) $(kA)^T = kA^T$, (iii) $(A+B)^T = A^T + B^T$, (iv) $(AB)^T = B^T A^T$
- A square matrix A is called
  (i) symmetric if $A^T = A$ and (ii) skew-symmetric if $A^T = -A$
- Any square matrix can be expressed as sum of a symmetric and skew-symmetric matrices.
- The diagonal entries of a skew-symmetric must be zero.
- For any square matrix A with real entries, $A + A^T$ is symmetric and $A - A^T$ is skew-symmetric and further $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$.
- Determinant is defined only for square matrices.
- $|A^T| = |A|$.
- $|AB| = |A|\,|B|$ where A and B are square matrices of same order.
- If $A = [a_{ij}]_{m\times n}$, then $|kA| = k^n |A|$, where k is a scalar.
- A determinant of a square matrix A is the sum of products of elements of any row (or column) with its corresponding cofactors; for instance, $|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$.
- If the elements of a row or column is multiplied by the cofactors of another row or column, then their sum is zero; for example, $a_{11}A_{13} + a_{12}A_{23} + a_{13}A_{33} = 0$.
- The determinant value remains unchanged if all its rows are interchanged by its columns.
- If all the elements of a row or a column are zero, then the determinant is zero.
- If any two rows or columns are interchanged, then the determinant changes its sign.
- If any two rows or columns are identical or proportional, then the determinant is zero.
- If each element of a row or a column is multiplied by constant k, then determinant gets multiplied by k.
- If each element in any row (column) is the sum of r terms, then the determinant can be expressed as the sum of r determinants.
- A determinant remains unaltered under a row (R_i) operation of the form $R_i + \alpha R_j + \beta R_k (j, k \neq i)$ or a Column (C_i) operation of the form $C_i + \alpha C_j + \beta C_k (j, k \neq i)$ where α, β are scalars.
- **Factor theorem :** If each element of $|A|$ is a polynomial in x and if $|A|$ vanishes for $x = a$, then $x - a$ is a factor of $|A|$.
- Area of the triangle with vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3) is given by the absolute value of $\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$.
  If the area is zero, then the three points are collinear.
- A square matrix A is said to be singular if $|A| = 0$ and non-singular if $|A| \neq 0$.

07_11th-Maths-Vol2-Unit7.indd 44 16-12-2022 11:00:11